

Business Requirements Document (BRD)

1. Document Control

Version	Date	Author	Description
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2. Introduction

Purpose	The application is designed to help individuals live an affordable, comfortable (debt-free) lifestyle while promoting financial literacy and growth by managing and tracking user expenditure, analyzing shopping lists, and suggesting more affordable alternatives. Emergency account access is available when needed to ensure both discipline and flexibility.
Project Overview	The application is designed for personal finance management, not institutional or governmental accounting. It supports users in limiting their spendings, making smarter shopping decisions and building long-term financial habits through automation, insights, and emergency override logic.
Scope	In Scope: Personal budgeting, account locking and planning, Emergency unlock verification, Expense management (including tracking, categorization and financial growth dashboards), Compliance and Audit, Integration with other systems (banking APIs), Analytics and Decision Support (analysis of shopping lists and recommendations), (IT management). Out of Scope: Functional areas Technical/infrastructure areas, Data scope

	Limitations, Government accounting and public sector reporting, Debt management for institutions, multi-year budget planning for ministries, compliance and audits with public finance laws.
Constraints	• Must meet personal financial data protection and security requirements. • Need internet connectivity, secure cloud hosting and exact decimal-based computations. • Project budget capped at around \$10,000 for MVP development. • Small team development within 20 weeks. • Will be dependent on stable APIs, emergency verification logic and reliable cloud services

3. Stakeholders

Name	Role	Responsibilities
Project Team (Developers / Designers)	System Creators	Design, develop, and test the system; ascertain functionality and usability.
Project Supervisor (Instructor/Mentor)	Evaluator	Oversee the project team, offer feedback, and evaluate progress.
End Users (Students, young adults, or individuals managing personal budgets)	System Users	Use the system to track expenses and improve financial literacy.
System Administrator	System Operator	Manage user accounts, monitor storage of data and optimize system performance.
Data Analyst	Data Processor	Process anonymous data for creating financial insights.
Cloud Service Provider	Technical Support	Provide Secure, reliable hosting and Application Programming Interface Services.
Compliance Authority (Finance / Data Regulators)	Legal Oversight	Ensure the system adheres to financial and data protection regulations.

UX Designer	Experience Architect	Designs interactive app interfaces and ensures accessibility.
Emergency verification Partner	External Validator	Confirms emergency unlock requests using location, receipts or urgency scoring.

4. Business Objectives

Objective	Success Metric
Financial risk control and management	Protects assets and enforces compliance.
Operational efficiency	Serves as a Time Saver through automating and simplifying financial activities.
Cash and liquidity management	Serves as a Cash Manager by optimizing working capital, its core objective is to ensure fiscal discipline (staying within budget), strategically allocate resources according to policy priorities, achieve operational efficiency (getting value for money), and reducing impulsive spending and encourage saving.
Strategic Insight and Decision Support	Provides Predictor capabilities for forward-looking strategy.
Emergency flexibility	Unlocks verified emergency requests.

5. Functional Requirements

5.1 Module 1: User access and account control

FR #1	Description
Functional Requirement	The System must allow users to log in using secure credentials(email/phone) with multi-factor authentication
Business Rule	1. The system must enforce MFA, encrypt credentials and log failed login attempts. 2. The system should allow biometric login (fingerprint/face ID).

FR #2	Description
Functional Requirement	The System must allow users to lock their accounts to restrict spending when budgets are exceeded
Business Rule	1. The system must restrict non-essential transactions while allowing essential bills. 2. The system should provide customizable lock rules (e.g. by category or amount)

FR #3	Description
Functional Requirement	The system must allow users to request emergency unlocks through the app during emergency situations
Business Rule	1. The system must require verification and log all unlock actions in an audit trail. 2. The system should allow urgency scoring or supporting documents.

FR #4	Description
Functional Requirement	The system must enforce different permissions for admins, users and auditors
Business Rule	1. The system must assign permissions based on role. 2. The system should notify admins of suspicious activity.

FR #5	Description
Functional Requirement	The system must manage user sessions securely to prevent unauthorized access
Business Rule	1. The system must auto logout for inactive users. 2. The system should allow configurable timeout settings.

5.2 Module 2: Expense tracking and categorization

FR #6	Description
Functional Requirement	The System must record all user transactions instantly and ensure proper classification
Business Rule	1. The system must timestamp and store transactions within 24 hours. 2. The system should use AI to auto-categorize transactions and allow manual overrides.

FR #7	Description
Functional Requirement	The System must allow users to view categorized spending summaries with historical comparisons
Business Rule	1. The system must update data daily, tag metadata and support filtering. 2. The system should allow export to CSV/PDF and provide dashboards.

FR #8	Description
Functional Requirement	The system must identify and prevent duplicate entries

Business Rule	1. The system must flag duplicates before saving. 2. The system should suggest merging or correcting duplicates.
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FR #9	Description
Functional Requirement	The system should support transactions in multiple currencies
Business Rule	1. The system must record local currency and convert it to a base currency. 2. The system should allow users to select preferred currency reports.

5.3 Module 3: Shopping list or similar items analysis and recommendations

FR #10	Description
Functional Requirement	The System must allow users to upload or enter shopping lists or its equivalent for analysis
Business Rule	1. The system must validate entries (price/vendor) before saving. 2. The system should auto-suggest missing details.

FR #11	Description
Functional Requirement	The System must suggest cheaper alternatives based on user preferences
Business Rule	1. The system must base recommendations on verified vendor data, preferences and location. 2. The system should track version history of recommendations.

FR #12	Description
Functional Requirement	The system must ensure vendors meet quality and security standards
Business Rule	1. The system must recommend only verified vendors. 2. The system should display vendor ratings and reviews.

FR #13	Description
Functional Requirement	The system should filter recommendations by eco-friendly or ethical vendors

Business Rule	1. The system must allow filtering by sustainability criteria. 2. The system should highlight cost vs. sustainability of trade-offs.
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5.4 Module 4: Financial growth dashboard

FR #14	Description
Functional Requirement	The System must allow users to view savings, spending trends and goals
Business Rule	1. The system must refresh dashboards weekly, allow drill-down and archive historical data. 2. The system should provide customizable goal widgets.

FR #15	Description
Functional Requirement	The System must generate monthly reports automatically.
Business Rule	1. The system must deliver reports within 5 days of month end, including savings rate, breakdown and recommendations. 2. The system should allow personalized layouts and export formats.

FR #16	Description
Functional Requirement	The system must warn users when spending exceeds the budget
Business Rule	1. The system must notify users of overspending. 2. The system should suggest rebalancing strategies.

FR #17	Description
Functional Requirement	The system should allow users to compare their spending against anonymized peer averages.
Business Rule	1. The system must provide benchmarking data. 2. The system should offer lifestyle insights (e.g. food vs. travel).

5.5 Module 5: Compliance and audit

FR #18	Description
Functional Requirement	The System must retain financial records securely for compliance.
Business Rule	1. The system must retain records for at least 12 months.

	2. The system should allow secure archiving/deletion.
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FR #19	Description
Functional Requirement	The System must comply with financial and data protection regulations
Business Rule	1. The system must comply with GDPR and financial data standards. 2. The system should provide consent management for data sharing.

FR #20	Description
Functional Requirement	The system must log all users' actions for transparency
Business Rule	1. The system must record add/edit/delete actions. 2. The system should allow export of logs for review.

5.6 Module 6: Risk Management

FR #21	Description
Functional Requirement	The System must detect unusual spending patterns to prevent fraud
Business Rule	1. The system must flag suspicious transactions. 2. The system should provide alerts and recommendations.

FR #22	Description
Functional Requirement	The System must track debts and arrears to help users avoid financial risks
Business Rule	1. The system must update arrears after every payment or new debt. 2. The system should provide alerts when arrears exceed user-defined thresholds.

5.7 Module 7: Integration with other systems

FR #23	Description
Functional Requirement	The System must connect securely with banks or e-wallets to retrieve transaction data
Business Rule	1. The system must comply with financial data security standards (e.g. PCI DSS).

	2. The system should allow users to link multiple accounts for consolidated tracking.
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FR #24	Description
Functional Requirement	The System must support importing and exporting financial data to common formats
Business Rule	1. The system must support CSV/XLSX for compatibility with accounting tools. 2. The system should allow scheduled exports for backups or analysis.

5.8 Module 8: Analytics and decision support

FR #25	Description
Functional Requirement	The System must provide dashboards for visualization
Business Rule	1. The system must refresh dashboards in real time. 2. The system should allow customizable chart types (bar, line, pie).

FR #26	Description
Functional Requirement	The System must generate financial forecasts based on user spending
Business Rule	1. The system must use at least the last 3 months of data for forecasts. 2. The system should provide personalized, accurate recommendations.

5.9 Module 9: IT Management

FR #27	Description
Functional Requirement	The System must ensure data is backed up and recoverable, store all financial records in a secure scalable database
Business Rule	1. The system must backup data daily. 2. The system should provide recovery options for users. 3. The system should be cloud based backup. 4. The system must encrypt data at rest and transit (AES-256).

6. Non-Functional Requirements

NFRs	Description
NFR #1: Performance	The System must support 1,000+ concurrent users.

NFR #2: Security	The System must encrypt all data.
NFR #3: Availability	The System must support uptime up to 99.9% especially during emergency unlocks
NFR #4: Usability	The System should support mobile interfaces and accessible.
NFR #5: Maintainability	The System should support modular architecture for easy updates and expanding features.

7. Risks & Issues

Risk	Mitigation
Misuse of emergency unlocks	Use multi-factor verification and urgency scoring
API failure	Monitor APIs and implement fallback logic
User frustration	Allow customizable lock settings and override options
Data breach	Use encryption, secure cloud hosting and access control
Inaccurate recommendations	Train AI models on user preferences and local pricing

8. Appendices (If applicable), Example: Glossary of Terms

Term	Definition
AES-256	Advanced Encryption Standard
APIs	Application Programming Interface
Data at rest and transit	Data stored on internal or external storage device and data in motion moving across a network
GDPR	General Data Protection Regulation
MFA	Multifactor Authentication
PCI DSS	A set of security standards for any organization that handles credit card information